

MAINTENANCE OPERATIONS CLIENT

Findings & Recommendations

Maintenance Performance & Operational Insights

Document Control	Detail
Engagement	Aug-Nov 2024
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Confirmed KPI	92% overall completion rate
Confirmed Reporting Impact	Approximately 3-4 hours of weekly manual reporting effort eliminated

Evidence rule: Findings below are retained from the engagement. Exact historical recommendations are not retained, so recommendation wording is presented as portfolio-proposed rather than claimed as the original client wording.

1. Executive Findings

Finding	Business Interpretation
92% overall completion rate	The analysis established the overall maintenance completion position. No historical target or benchmark is retained.
Backlog identified	Outstanding maintenance workload required management visibility.
Overdue work identified	Some incomplete work had passed its due date under the confirmed overdue rule.
Differences across teams	Performance varied across teams; exact team rates and rankings are not retained.
Workload imbalance	Maintenance workload was unevenly distributed across the operation.
High-priority pending work	Critical outstanding work was visible within the maintenance workload.
Recurring maintenance categories	Repeated maintenance categories were identified for operational review.
Equipment-related issues/patterns	Equipment-level analysis surfaced patterns requiring further operational attention.

2. Key Business Rules

Measure	Rule
Completion Rate	Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100
Overdue	Due Date < Current Date AND Status != Completed
Backlog	Confirmed analytical area; exact historical definition not retained

Important: The 92% completion rate was surfaced by the analysis. It must not be described as a 92% improvement or as an increase caused by the project.

3. Findings to Management Focus

Confirmed Finding	Management Focus
Backlog	Maintain visibility of outstanding work and its operational concentration.
Overdue work	Review late incomplete work separately from the overall completion KPI.
Team differences	Interpret team performance together with workload and priority context.
Workload imbalance	Review how work is distributed across teams and technicians.
High-priority pending work	Ensure critical outstanding work is visible for prioritization.
Recurring categories	Investigate repeated maintenance activity rather than treating each record as isolated.
Equipment patterns	Use equipment-level patterns as investigation signals, not automatic root-cause conclusions.

4. Portfolio-Proposed Recommendations

Status: The following actions are derived from the confirmed findings for portfolio purposes. They are not presented as the exact historical recommendations delivered to the client.

Finding	Portfolio-Proposed Recommendation	Expected Management Value
Overdue work	Use a dedicated overdue exception view based on the confirmed overdue rule and review it routinely.	Keeps late incomplete work visible.
High-priority pending work	Prioritize high-priority outstanding items within the exception-management view.	Supports attention to critical work.
Workload imbalance	Review workload alongside completion, overdue and priority before reallocating work.	Supports more informed resource decisions.
Team differences	Compare teams using governed measures and workload context rather than raw completion counts alone.	Improves fairness and interpretability of comparisons.
Recurring categories	Investigate recurring maintenance categories to determine whether repeat activity warrants process or preventive-maintenance review.	Supports root-cause investigation.
Equipment patterns	Review assets/equipment with repeated maintenance activity and overdue work for deeper operational assessment.	Supports targeted equipment investigation.
Backlog	Define and govern one backlog rule, then monitor the measure consistently in Power BI.	Creates consistent backlog visibility.

5. Reporting & Decision-Support Outcome

The Power BI solution consolidated maintenance reporting into a reusable operational view for the Maintenance Manager. It supported monitoring of completion, outstanding work, overdue items, priority, teams, technicians, categories and assets/equipment.

Confirmed efficiency outcome: Approximately 3-4 hours of weekly manual reporting effort were eliminated.

6. Outcome Boundaries

Supported	Not Supported
92% overall completion rate was observed.	A claim that completion improved to 92% because of the project.
Backlog and overdue work were identified.	Exact backlog or overdue counts/percentages.
Team differences and workload imbalance were identified.	Exact team rankings or causal explanations.
High-priority pending work was identified.	Exact number of high-priority pending items.
Recurring categories and equipment patterns were identified.	Exact category/asset rankings or root causes.
Reporting effort reduced by approximately 3-4 hours weekly.	Annualized cost savings or a named individual's time saving.

7. Final Project Value

The project combined data preparation, KPI definition, operational analysis and Power BI reporting to convert fragmented maintenance records into clearer management visibility. Its defensible value is the combination of the retained 92% completion insight, visibility into operational exceptions and approximately 3-4 hours of weekly manual reporting effort eliminated.